

Mixed quadratic and Lorentz constraints: implementation equations

Let $g(x)$ stack affine upper bounds, convex quadratic upper bounds $g_i(x) = \frac{1}{2}x^T Q_i x + q_i^T x - u_i$, and cone rows $-F_j x - f_j$. The model is $\min \frac{1}{2}x^T P x + c^T x$ subject to $Ax = b$ and $g(x) + s = 0$, with s, z in a product of positive rays and Lorentz cones. Write $J = g'(x)$ and $H = P + \sum_{i \in \text{quadratic}} z_i Q_i$. The original residuals are

$$r_d = Px + c + A^T y + J^T z, \quad r_e = Ax - b, \quad r_g = g(x) + s.$$

With proximal centers (ξ, η, ζ) , use $\bar{r}_d = r_d + \rho(x - \xi)$, $\bar{r}_e = r_e - \delta(y - \eta)$ and $\bar{r}_g = r_g - \delta(z - \zeta)$. The derivative has primal block $H + \rho I$ and negative dual regularization.

For a Lorentz cone use $e = (1, 0)$, $(a, u) \circ (b, v) = (ab + u^T v, av + bu)$ and the barrier $-\frac{1}{2} \log(a^2 - \|u\|^2)$. Each scalar inequality and each cone contributes one to ν . Set $\mu = s^T z / \nu$. The symmetric Nesterov–Todd scaling satisfies $\lambda = Wz = W^{-1}s$, with $W = W^T \succ 0$. For scalar rows $W = \sqrt{s/z}$. In scaled coordinates,

$$L_\lambda(W^{-1}\Delta s + W\Delta z) = r_c, \quad L_\lambda v = \lambda \circ v.$$

Thus $\Delta s + W^2 \Delta z = t$, where $t = WL_\lambda^{-1}r_c$. Eliminating slack gives

$$\begin{bmatrix} H + \rho I & A^T & J^T \\ A & -\delta I & 0 \\ J & 0 & -D \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} = \begin{bmatrix} -\bar{r}_d \\ -\bar{r}_e \\ -\bar{r}_g - t \end{bmatrix}, \quad D = W^2 + \delta I.$$

Set $B = D^{-1/2}$ and replace each cone Jacobian by BJ , its right-hand side by $-B\bar{r}_g - BW h$, and its dual step by $v = B^{-1}\Delta z$, where $h = L_\lambda^{-1}r_c$. The resulting dual block is $-I$. This congruence leaves the existing PIQP backend interface unchanged. Apply BW directly through its bounded eigenvalues. Recover the dual step as $\Delta z = Bv$ and the cone slack step from affine feasibility, $\Delta s = -\bar{r}_g - J\Delta x + \delta\Delta z$, to avoid cancellation.

The affine predictor uses $r_c = -\lambda \circ \lambda$. For its direction $(\Delta x_a, \Delta s_a, \Delta z_a)$, estimate μ_a using admissible cone steps and set $\sigma = \min(1, \max(0, \mu_a/\mu))^3$. The combined corrector uses

$$r_c = \sigma \mu e - \lambda \circ \lambda - (W^{-1}\Delta s_a) \circ (W\Delta z_a).$$

For native quadratics also subtract $\frac{1}{2}\Delta x_a^T Q_i \Delta x_a$ from the feasibility right-hand side and $\sum_i \Delta z_{a,i} Q_i \Delta x_a$ from the stationarity right-hand side. These are the exact second-order Taylor remainders of the corresponding unregularized residuals. They are not extra Hessian terms. Use one common step for primal and dual variables to retain these factors. Fraction-to-boundary keeps scalar pairs positive and cone pairs interior. Backtracking evaluates the actual nonlinear residuals with fixed centers, fixed target $\sigma\mu$, and fixed W . Its complementarity residual is $(W^{-1}s_{\text{trial}}) \circ (Wz_{\text{trial}}) - \sigma\mu e$. Reject a direction if it cannot reduce this residual merit, and retry a centered direction without affine remainder terms. This safeguard is an implementation choice, not a convergence theorem.

Equilibrate with $x = E\hat{x}$ and objective multiplier $\gamma > 0$. A scalar constraint scale $\alpha_i > 0$ gives $\hat{Q}_i = \alpha_i E^T Q_i E$, $\hat{q}_i = \alpha_i E^T q_i$, and $\hat{u}_i = \alpha_i u_i$. A cone uses one common positive scale β_j for all its rows. Return $s_i = \hat{s}_i/\alpha_i$ and $z_i = \alpha_i \hat{z}_i/\gamma$, with the analogous cone and equality maps. For a rotated cone use $T(a, b, v) = (a + b, a - b, \sqrt{2}v)$, return $s_r = T^{-1}s_o$, and return $z_r = T^T z_o$. The latter preserves the dual pairing.

Initialize affine rows and cone rows through PIQP’s regularized auxiliary linear system with unit slack scaling. Exclude native quadratic rows from that auxiliary objective, evaluate them at its primal point, then choose positive slack-dual pairs. Repair Lorentz vectors by shifting their first coordinate. Allocate structural support before initialization.

Termination uses original-coordinate stationarity, affine feasibility, true quadratic feasibility, slack consistency, primal and dual cone membership, and complementarity. No infeasibility certificate is inferred from proximal stagnation. A complementarity sum at an infeasible point is not an objective gap. Writing the Lagrangian as $\frac{1}{2}x^T H x + \ell^T x + k$, for singular H the exact dual objective requires $\ell \in \text{range}(H)$ and is $k - \frac{1}{2}\ell^T H^\dagger \ell$; otherwise it is $-\infty$. The implementation reports the primal objective and KKT residuals without claiming this expensive dual evaluation.